

Some Potentially Useful Formulas and Definitions

- $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$

- Make sure you explain the reasoning process behind your conclusion by referring to the (1) observed statistic, (2) p-value (3) justify strength of evidence against the null model (very strong, strong, moderate, weak) based on the p-value, (4) reject/fail to reject the null hypothesis, and (5) overall conclusion in context. Make sure you refer to the parameter in your conclusion.

p-value	Strength of Evidence
$0.10 < p\text{-value}$	Not much evidence against the null hypothesis; null is plausible
$0.05 < p\text{-value} \leq 0.10$	Moderate evidence against the null hypothesis
$0.01 < p\text{-value} \leq 0.05$	Strong evidence against the null hypothesis
$p\text{-value} \leq 0.01$	Very strong evidence against the null hypothesis

Table 1: Guidelines for evaluating the strength of evidence from a p-value

You may use the remainder of this page for scratch work; please keep in mind that NOTHING written here will be graded.